

BARKAM, ChinaWMO: **561720**Lat: **31.8991N**Lon: **102.2289E**Elev: **2666**StdP: **73.14**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-9.6	-8.1	-21.0	0.8	3.2	-19.1	1.0	1.6	5.7	6.1	4.5	6.2	0.3	90	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	29.1	15.6	27.2	15.1	25.5	14.5	17.4	24.7	16.7	23.6	16.1	22.5	2.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	15.4	19.1	14.9	14.7	18.4	14.3	14.2	17.9	61.3	24.8	58.7	23.6	56.3	22.7	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.3	3.5	-12.8	32.7	1.1	1.2	-13.6	33.5	-14.3	34.2	-14.9	34.9	-15.7	35.8		
(5)			-14.0	19.0	1.0	1.0	-14.6	19.7	-15.2	20.3	-15.7	20.8	-16.4	21.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	1.1	4.6	7.6	11.1	14.1	16.4	18.0	18.0	15.5	10.7	5.3	0.9
	DBStd	6.56	2.35	2.54	2.54	2.73	2.77	2.59	2.36	2.52	2.90	2.82	2.29	2.20
	HDD10.0	978	277	150	80	19	3	0	0	0	1	24	142	283
	HDD18.3	3001	535	383	331	216	134	68	36	37	92	236	391	542
	CDD10.0	1088	0	0	7	53	129	192	247	247	166	47	0	0
	CDD18.3	69	0	0	0	0	2	10	24	26	7	0	0	0
	CDH23.3	926	0	0	1	22	87	119	266	300	119	12	0	0
	CDH26.7	209	0	0	0	2	15	21	66	81	24	1	0	0
Wind	WSAvg	1.1	1.0	1.2	1.4	1.4	1.3	1.0	1.0	1.0	1.0	0.9	0.9	0.9
Precipitation	PrecAvg	769	2	6	19	46	102	154	134	105	137	55	7	2
	PrecMax	940	11	15	43	98	163	270	257	166	220	155	44	8
	PrecMin	632	0	0	4	20	38	86	62	54	44	15	0	0
	PrecStd	82	3	4	10	17	28	44	45	29	38	29	9	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	19.1	22.2	25.9	28.7	29.1	31.0	31.3	29.5	25.1	17.2	12.4
		MCWB	2.5	5.0	7.0	10.2	13.5	15.8	16.3	16.2	15.6	13.6	6.7	2.8
	2%	DB	11.7	16.5	19.4	23.3	25.7	26.4	28.5	29.0	26.7	21.4	15.5	10.5
		MCWB	1.6	3.8	5.9	8.9	12.0	15.2	16.0	15.9	15.1	11.7	5.7	1.7
	5%	DB	10.0	14.3	17.4	21.1	23.5	24.3	26.4	26.8	24.3	19.1	13.8	9.1
		MCWB	0.8	3.0	5.2	8.1	11.4	14.6	15.7	15.4	14.4	10.4	4.7	0.8
	10%	DB	8.2	12.3	15.4	18.9	21.2	22.3	24.1	24.5	21.8	16.9	12.0	7.6
		MCWB	-0.1	2.2	4.6	7.5	10.8	14.1	15.3	15.0	13.6	9.4	3.9	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	5.5	7.9	11.3	15.3	18.0	18.4	18.1	17.1	14.3	7.7	3.5
		MCDB	12.4	16.8	18.1	22.9	25.4	25.1	25.4	26.1	25.9	23.2	15.5	10.9
	2%	WB	2.0	4.3	6.7	10.0	13.7	16.8	17.5	17.1	16.0	12.7	6.5	2.2
		MCDB	10.8	14.6	16.7	19.5	22.0	23.1	24.4	25.1	23.8	19.7	13.6	9.5
	5%	WB	1.0	3.4	5.9	9.2	12.6	15.9	16.8	16.3	15.1	11.4	5.5	1.2
		MCDB	9.3	12.9	15.3	18.0	20.0	21.8	23.2	23.6	21.9	17.0	12.1	8.2
	10%	WB	0.1	2.5	5.1	8.4	11.7	15.0	16.2	15.6	14.2	10.2	4.6	0.3
		MCDB	7.6	11.1	13.5	16.6	18.3	20.1	21.9	22.1	19.9	15.0	10.6	6.8
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.7	15.5	15.1	14.1	12.1	12.7	13.4	13.0	13.1	14.8	14.8
		MCDBR	18.0	20.2	19.3	19.5	18.3	16.0	17.4	17.9	17.3	16.5	16.8	15.9
		MCWBR	9.8	10.0	8.3	7.4	6.5	5.7	6.2	6.0	6.6	7.3	8.3	8.8
	5% WB	MCDDBR	17.1	18.9	16.9	16.0	15.1	13.2	13.0	13.8	14.2	14.5	15.3	15.1
		MCWBR	9.5	9.3	7.2	6.1	5.7	5.1	5.2	5.2	5.6	6.4	7.7	8.5
Clear-Sky Solar Irradiance	taub		0.275	0.316	0.397	0.421	0.415	0.401	0.414	0.421	0.375	0.319	0.277	0.256
	taud		2.443	2.315	2.046	2.036	2.154	2.293	2.274	2.232	2.357	2.469	2.505	2.526
	Ebn at Noon		958	940	878	870	876	884	870	861	892	927	950	968
	Edh at Noon		91	114	162	171	153	133	135	139	117	97	85	79
All-Sky Solar Radiation	RadAvg		3.84	4.54	5.37	6.06	5.88	4.83	5.15	4.88	4.32	3.94	3.84	3.69
	RadStd		0.28	0.26	0.39	0.41	0.43	0.51	0.51	0.59	0.36	0.26	0.20	0.20